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California needs a million EV charging stations — but that's 'unlikely' and 'unrealistic'

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California will have to build public charging stations at an unprecedented — and some experts say unrealistic — pace to meet the needs of the 7 million electric cars expected on its roads in less than seven years.

The sheer scale of the buildout has alarmed many experts and lawmakers, who fear that the state won't be prepared as Californians purchase more electric cars.

A million public chargers are needed in California by the end of 2030, according to the state's projections — almost 10 times more than the number available to drivers in December. To meet that target, 129,000 new stations — more than seven times the current pace — must be built every year for the next seven years. Then the pace would have to accelerate again to reach a target of 2.1 million chargers in 2035.

A robust network of public chargers — akin to the state's more than 8,000 gas stations — is essential to ensure that drivers will have the confidence to purchase electric vehicles over the next several years.

"It is very unlikely that we will hit our goals, and to be completely frank, the EV goals are a noble aspiration, but unrealistic," said Stanford professor Bruce Cain, who co-authored a policy briefing detailing California's electric vehicle charging problems. "This is a wakeup call that we address potential institutional and policy obstacles more seriously before we commit blindly."

Under California's landmark electric car mandate, a pillar of Gov. Gavin Newsom's climate change agenda, 68% of all new 2030 model cars sold in the state must be zero emissions, increasing to 100% for 2035, when 15 million electric cars are expected in California.

"We're going to look really silly if we are telling people that they can only buy electric vehicles, and we don't have the charging infrastructure to support that," said Assemblymember Jesse Gabriel, a Democrat from Encino who introduced a package of unsuccessful bills last year aimed at expanding access to car chargers.

"We are way behind where we need to be," Gabriel told CalMatters.

Big obstacles stand in the way of amping up the pace of new charging stations in public places. California will need billions of dollars in state, federal and private investments, streamlined city and county permitting processes, major power grid upgrades and accelerated efforts by utilities to connect chargers to the grid.

State officials also are tasked with ensuring that charging stations are available statewide, in rural and less-affluent areas where private companies are reluctant to invest, and that they are reliable and functioning whenever drivers pull up.

In Pacific Gas & Electric's vast service area, home to 40% of all Californians, electric car purchases are moving twice as fast as the buildout of charging stations, said Lydia Krefta, the utility's director of clean energy transportation. Californians now own more than 1.5 million battery-powered cars.

Patty Monahan, who's on the Energy Commission, the state agency responsible for funding and guiding the ramp-up, told CalMatters that she is confident that California can build the chargers its residents need in time.

The agency's estimate of the current chargers is likely an undercount, she said. In addition, fast-charging stations could play a bigger role than initially projected, meaning hundreds of thousands of fewer chargers might be

needed. Also, as the ranges and charging speeds on cars improve, there may be less demand for public chargers.

"California has a history of defying the odds," Monahan said. "We have a history of advancing clean cars, clean energy, writ-large. We have naysayers left and right saying you can't do it, and then we do it."

Barriers to private investments: an uncertain market

On a September day last year, Monahan spoke behind a podium in the parking lot of a Bay Area grocery store. A row of newly constructed car chargers rose behind her.

"Let's celebrate for a moment," she said.

California had met its goal of 10,000 fast electric chargers statewide — two years ahead of a target set in 2018.

Fast chargers like the new ones at the grocery store are increasingly seen as critical to meeting the needs of drivers. They can power a car to 80% in 20 minutes to an hour, while the typical charger in use today, a slower Level 2, takes from four to 10 hours.

But installing and operating fast chargers is an expensive business — one that doesn't easily turn a profit.

Nationwide each fast charger can cost up to $117,000, according to a 2023 study. And in California, it could be even more — between $122,000 and $440,000 each, according to a separate study, although the Energy Commission said the range was $110,000 to $125,000 for one of its programs.

Most of America's publicly traded charger companies have been forced to seek more financing, lay off workers and slow their network build outs, analysts said. EVgo, for instance, has seen its share price crater, as has ChargePoint, which specializes in selling the slower, Level 2 hardware.

California stands apart from other states — it has by far the most chargers and electric car sales, and more incentives and policies encouraging them.

Tesla, America's top-selling electric car manufacturer, dominates fast-charging in both California and the U.S. — but the company didn't get into the business to sell charges to drivers; it got into the charger business to sell its electric cars. Initially Tesla Superchargers were exclusive to its drivers, but starting this year other EV drivers can use them after Tesla provided ports to Ford and other automakers.

Tesla's manufacturing prowess, supply chain dominance and decade-plus of experience with fast chargers have given it an edge over competitors — a coterie of unprofitable, publicly traded startups, as well as private companies that often benefit from public subsidies, according to analysts.

"All the automakers joined forces with their biggest competitor," said Loren McDonald, chief executive of the consulting firm EVAdoption. "If that doesn't tell you how bad fast-charging networks and infrastructure were, I don't know what else does."

First: An EVgo fast charger at El Mercado Plaza Shopping Center in Union City. Last: A Hyundai Ioniq 6 charges at an Electrify America station in Oakland. Photos by Loren Elliott and Camille Cohen for CalMatters

Now Tesla is showing uncertainty about the future of its charging business amid slumping car sales, and eliminated nearly its entire 500-member Supercharger team in April. Then chief executive Elon Musk said in May that he would spend $500 million to expand the network and hired back some fired workers.

In California, Electrify America, a privately held company, was created by Volkswagen as a settlement for cheating on emissions tests for its gas-powered cars. The company is spending $800 million on California chargers, building a robust network of 260 stations, with more than half in low-income communities, including the state's worst

charging desert, Imperial County.

The problem is Electrify America was ranked dead last in a consumer survey last year, and its chargers have been plagued by reliability problems and customer complaints. The California Air Resources Board in January directed Electrify America to "strive to achieve charger reliability consistent with the state of the industry." A company spokesperson said the dissatisfaction showed "an industry in its growth trajectory." There are signs of improvement, based on consumer data from the first three months of this year.

Startups continue to jump into the charging business, with the number of companies offering fast chargers growing from 14 in 2020 to 41 in 2024, EVAdoption said. Seven carmakers formed a $1 billion venture to build a 30,000-charger network in North America. And gas stations such as Circle K are offering more charging because electric car customers spend more time shopping while waiting for their rides to juice up.

But the realization that charging is a costly business has set in on Wall Street, and that doesn't seem likely to change anytime soon. "Can public EV fast-charging stations be profitable in the United States?" the consultancy McKinsey & Company asked.

"The fervor, the excitement from the investor base, has definitely dwindled quite a bit, given the prospects that EV adoption in the U.S. is going to be slower, revenue growth is really slower, the path to profitability is going to be slower, and they might need more capital than everyone originally expected," said Christopher Dendrinos, a financial analyst who covers electric car charging companies for the investment bank RBC Capital Markets.

The stakes are high for California when it comes to encouraging investments in expensive fast chargers: If 63,000 additional ones were built, California might need 402,000 fewer slower Level 2 chargers in 2030, according to an alternative forecast by the Energy Commission.

Billions of public dollars: Will it be enough?

Nationwide $53 billion to $127 billion in private investments and public funding is needed by 2030 to build chargers for about 33 million electric cars, according to a federal estimate. Of that, about half would be for public chargers.

Congress and the Biden administration have set aside $5 billion for a national network of fast chargers. So far only 33 in eight locations have been built, but more than 14,000 others are in the works, according to the Federal Highway Administration. California's share of the federal money totals $384 million; about 500 fast chargers will be built with an initial $40.5 million, said Energy Commission spokesperson Lindsay Buckley.

In addition, the state has spent $584 million to build more than 33,000 electric car chargers through its Clean Transportation Program, funded by fees drivers pay when they register cars. The Legislature extended that program for an additional decade last year.

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Jesus Gonzalez, 25, Los Angeles

Housing: Rented house

Car model: Mustang Mach-E

Charging usage: Public chargers only

Gonzalez lives in the dense Lincoln Heights neighborhood of Los Angeles. He bought his electric car when his older, gas-powered car started giving him problems in 2022.

Gonzalez soon realized charging his car was more complicated than he had anticipated. He initially relied on fast

chargers, utilizing the free kilowatts provided by Electrify America with his car purchase. Those credits lasted him only a couple months.

These days he drives 10 to 15 minutes to the nearest Electrify America station, uses a free fast charger run by his local utility or a couple of nearby slower Level 2 chargers.

"At first, the time spent charging was okay," he said. But these days, "there are longer lines. So instead of just having to wait 20 minutes charging, you have to wait maybe 30 minutes for all the cars that are charging to clear up, and then the time for you to charge your car. I think that's the biggest pain point."

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Newsom has committed to spending $1 billion through 2028 on chargers with his "California Climate Commitment," Buckley said. But this year Newsom and the Legislature trimmed $167 million from the charger budget as the state faces a record deficit. A lobbyist for the Electric Vehicle Charging Association said "the state pullback sends a very challenging message" to the industry.

California's commitment to charger funding is "solid," despite the cuts, Buckley said. They have not yet estimated the total investment needed in California to meet the targets.

But Ted Lamm, a UC Berkeley Law researcher who studies electric car infrastructure, said the magnitude of building what California needs in coming years likely dwarfs the public funding available.

State and federal programs will "only fund a fraction," and the state needs to spend that money on lower-income communities, he said.

Another possible funding source is California's Low Carbon Fuel Standard, which is expected to be revised in November. The program requires carbon-intensive fuel companies to pay for cleaner-burning transportation. Utilities get credits and use that money to pay for chargers, rebates to car buyers and grid improvements, said Laura Renger, executive director of the California Electric Transportation Coalition, which represents utilities.

"I think with that, we would have enough money," Renger said. She said the program's overhaul could help utilities invest "billions" in chargers and other electric car programs over the next two decades.

Backlogged local permits and grid delays

One of the biggest barriers to more chargers isn't money. It's that cities and counties are slow to approve plans for the vast number of stations needed.

State officials only have so much political power to compel local jurisdictions to do what they want — a reality made abundantly clear by the housing crisis, for instance. California relies on grants and persuasion to accomplish its goals, and the slow buildout of chargers shows how those strategies can fall short, said Stanford's Cain.

"The locals cannot be compelled by regulatory agencies to make land and resources available for what the state wants to achieve," Cain said.

The same obstacles have marked the state's broader effort to electrify California and switch to clean energy. Local opposition and environmental reviews sometimes hold up large solar projects and transmission projects for years.

California has created a "culture of regulation that emphasizes the need to be extra careful and extra perfect, but this takes an incredible amount of time," Steve Bohlen, senior director of government affairs at Lawrence Livermore National Laboratory, said last month at the inaugural hearing of the state Assembly's Select Committee on Permitting Reform.

"We're moving into a period of rapid change, and so perfect can't be the enemy of the good."

First: Workers install a transformer to power electric car chargers in Calexico. Last: A row of new public chargers there. Photos by Adriana Heldiz, CalMatters

Chargers aren't as complicated as large-scale solar or offshore wind projects. But most chargers installed in public spaces do need a land-use or encroachment permit, among other approvals. California has passed laws requiring local jurisdictions to streamline permits for chargers. What's more, the Governor's Office of Business Development now grades cities and counties using a scorecard and maintains a map displaying who has, or hasn't, made life easier for car charger builders. But these strategies only go so far.

"It doesn't matter how many requirements you put on (local governments)," Lamm said. "If they just don't have the time in the day to do it … it's going to sit in the backlog, because that's how it works."

The delays have consequences. Getting a station permitted in California, on average, takes 26% longer than the national average, Electrify America reported. Designing and constructing a station in California can cost on average 37% more than in other states because of delays in permitting and grid connections. A utility on average takes 17 weeks after work is completed to connect chargers to the grid, Electric America said.

Powering large charging projects often requires grid upgrades, which can take a year or more for approval, said Chanel Parson, a director at Southern California Edison. Supply chain issues also make getting the right equipment a challenge.

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Andrew Krulewitz, 37, Oakland

Housing: Single family home

Car model: Hyundai Ioniq 6

Charging usage: Home chargers and public chargers

When he lived in San Francisco, Krulewtiz had access to a Tesla through his job, and relied entirely on Level 2 chargers near his home, one of which was free.

"I had no parking, I parked on the street, and I had to find places to charge," he said. He would also use Tesla's fast-charging Supercharger network. "When talking about infrastructure, it's (Tesla) Supercharging and everybody else."

He later moved to Oakland and drove an electric Fiat. He now owns a home, but when he lived in an apartment, he had to run a cord from his apartment building onto the street where he parked his car.

A Level 1 charger, like those found in many homes, adds about 4 miles of driving range per hour of charge. "I think Level 1 is a totally unsung solution to most people's needs, but it's not ideal if you need to charge every night," he said.

"It used to be that public charging was more expensive than home charging, and now that has flipped — home charging can be more expensive than public charging."

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Edison, which has a 10-year plan to meet expected demand, has asked the utilities commission for approval to upgrade the grid where it anticipates high charging demand.

"Every EV charging infrastructure project is a major construction project," Parson said. "There are a number of variables that influence how long it takes to complete the project."

Impatient with broken chargers, bad service

Inspired to help the nation reduce its dependence on fossil fuels, Zach Schiff-Abrams of Los Angeles bought a Genesis GV60. As a renter, he has relied on public charging, primarily using Electrify America stations — and that's been his biggest problem about owning an electric car.

Charging speeds have been inconsistent, he said, with half-hour sessions providing only a 15 to 30% charge, and he often encounters broken chargers.

"I believe in electrical, so I'm really actually trying to be a responsible consumer," Schiff-Abrams said. "I want to report them when they're down, but the customer service is horrible."

Electric vehicles line up at a busy Electrify America charging station in Kettleman City. Photo by Larry Valenzuela, CalMatters/CatchLight Local

For years, the reliability of charging networks has been a well-documented problem. Only 73% of fast chargers in the San Francisco Bay Area were functional in a 2022 study. The growth of the EV market has put increasing strain on public charging stations, a consumer survey found.

In January, the California Air Resources Board approved a final $200 million spending plan for Electrify America — but not before board chair Liane Randolph scolded its CEO.

Randolph — arguably one of America's top climate regulators — told CEO Robert Barrosa about an exchange she had with his company's customer service line after finding a broken charger at a station along Interstate-5.

"It didn't work," Randolph said during the board meeting. "Called the customer service line, waited like 10-ish minutes. ...(The charger) was showing operable on the app and the guy goes, 'oh, my data is showing me that it has not had a successful charge in three days.'"

"These issues are not easy," Barrosa responded. "Our head is not in the sand," he told board members earlier. "We are listening to customers."

But Randolph, addressing journalists at a conference in Philadelphia, pushed back against the idea that because the transition to electric vehicles is happening gradually that it's a failure. Many people will rely on charging at home or work, and batteries are becoming more efficient.

"The infrastructure is continuing to be rolled out at a rapid pace," Randolph said. "It doesn't all have to be perfect instantly. It's a process. And it's a process that's continuing to move."

Data journalists Erica Yee and Arfa Momin contributed to this report.

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